Earth's inner core is made up of two transition metals: **iron and nickel.**

Transition metals

What we usually think of as "metals" mostly belong to the group of elements known as transition metals. Most are hard and shiny. They have many other properties in common, including high boiling points and being good at conducting heat and electricity.

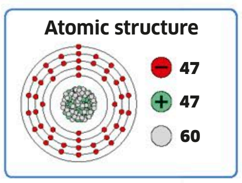
The transition metals make up the biggest element block in the periodic table, spreading out from group 3 through to group 12, and across four periods (see pp.28–29). This wide spread indicates that, although they are similar in many ways, they vary in others, such as how easily they react and what kinds of compounds they form.

Some of these metals have been known for more than 5,000 years. Some were only discovered in the 20th century. This is a selection of some of the 38 transition metals.

SILVER

Argentum

Discovered: c. 3000 BCE



47 107.87
Ag
SILVER

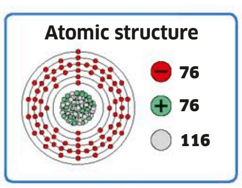
Like gold and copper, silver was one of the elements known and used by the earliest civilizations. It is valuable and easy to mold and used to be made into coins. Today, coins are made of alloys (see pp.62–63). Silver is still one of the most popular metals and is used for jewelry and decorative objects.

Chunk of silver
Silver metal reacts with the sulfur in air, which produces a black coating. That is why silver needs polishing to stay shiny.

OSMIUM

Osmium

Discovered: 1803



76 190.23
Os
OSMIUM

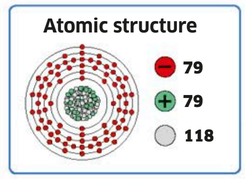
This rare, blue-shimmering metal is incredibly dense—a tennis ball-sized lump of osmium would have a mass of 7.7 lb (3.5 kg). If exposed to air, it reacts with oxygen to form a poisonous oxide compound, so for safe use it needs to be combined with other metals or elements. The powder used to detect fingerprints contains osmium.

Hard but brittle
This sample of refined osmium looks solid enough, but the tiny cracks all over it show that it is fragile in its pure form.

GOLD

Aurum

Discovered: c. 3000 BCE



79 196.97
Au
GOLD

Since ancient times, gold has been treasured because of its great beauty, and also because it doesn't get damaged by corrosion—it keeps its yellow sheen and does not rust. Easy to shape, it can be seen in jewelry, Egyptian masks, building decorations, and also in electronics. It doesn't easily react or form compounds with other elements.

Gold nugget
In nature, pure gold can be found in nuggets such as this or, more commonly, as grains inside rocks.